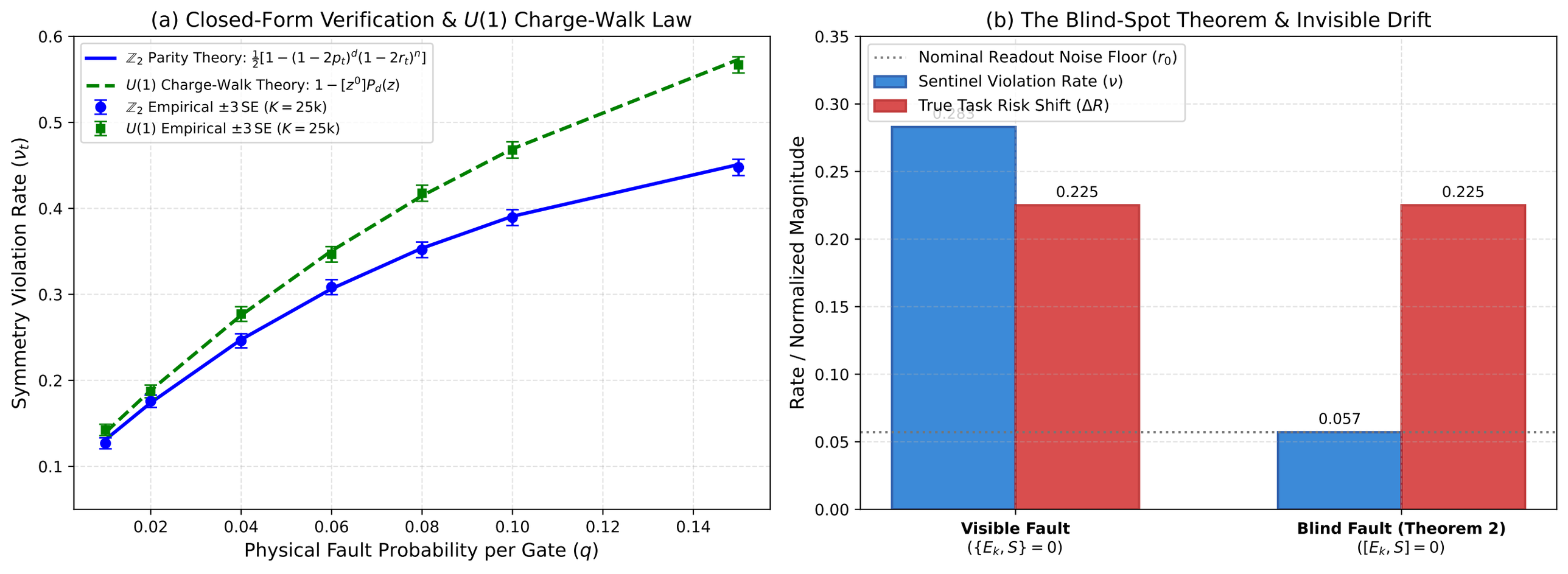
(a) Closed-Form Verification & $U(1)$ Charge-Walk Law
$\mathbb{Z}_2$ Parity Theory: $\frac{1}{2}[1-(1-2p_t)^d(1-2r_t)^n]$
$U(1)$ Charge-Walk Theory: $1-[z^0]P_d(z)$
$\mathbb{Z}_2$ Empirical ± 3 SE ($K = 25$k)
$U(1)$ Empirical ± 3 SE ($K = 25$k)
Symmetry Violation Rate (ν_t)
Physical Fault Probability per Gate (q)
0.1
0.2
0.3
0.4
0.5
0.6
0.02
0.04
0.06
0.08
0.10
0.12
0.14
(b) The Blind-Spot Theorem & Invisible Drift
Nominal Readout Noise Floor (r_0)
Sentinel Violation Rate (ν)
True Task Risk Shift (ΔR)
Rate / Normalized Magnitude
0.283
0.225
0.057
0.225
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Visible Fault
($\{E_k, S\} = 0$)
Blind Fault (Theorem 2)
($[E_k, S] = 0$)